

confidence interval for the mean

what if σ is unknown? $\Rightarrow$ STUDENT'S-t DISTRIBUTION

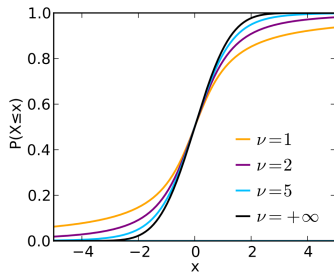
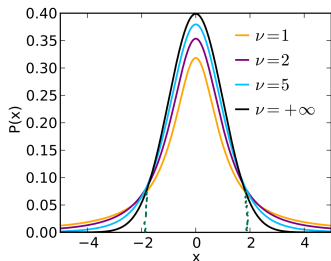
- $t_n = \frac{Z}{\frac{V}{\sqrt{n}}}$ with $\begin{cases} Z \approx N(x) \\ V^2 \approx \chi_n^2 \end{cases}$, and $\nu = n$ are the degrees of freedom

- symmetric around the mean 0 like $N(x)$ $\Rightarrow$ MEDIAN = MODE

- "shorter" and "fatter" than $N(x)$ $\Rightarrow$ LARGER SPREAD (THAN $N(x)$)

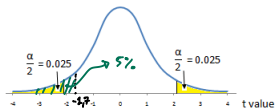
- tends to $N(x)$ when ν grows

- table of values available for the cumulative distribution



Student's t Distribution Table

For example, the t value for
18 degrees of freedom
is 2.101 for 95% confidence
interval (2-Tail $\alpha = 0.05$).



	90%	95%	97.5%	99%	99.5%	99.95%	1-Tail Confidence Level
	80%	90%	95%	98%	99%	99.9%	2-Tail Confidence Level
	0.100	0.050	0.025	0.010	0.005	0.0005	1-Tail Alpha
df	0.20	0.10	0.05	0.02	0.01	0.001	2-Tail Alpha
1	3.0777	6.3138	12.7062	31.8205	63.6567	636.6192	
2	1.8856	2.9200	4.3027	6.9646	9.9248	31.5991	
3	1.6377	2.3534	3.1824	4.5407	5.8409	12.9240	
4	1.5332	2.1318	2.7764	3.7469	4.6041	8.6103	
5	1.4759	2.0150	2.5706	3.3649	4.0321	6.8688	
6	1.4398	1.9432	2.4469	3.1427	3.7074	5.9588	
7	1.4149	1.8946	2.3646	2.9980	3.4995	5.4079	
8	1.3968	1.8595	2.3060	2.8965	3.3554	5.0413	
9	1.3830	1.8331	2.2622	2.8214	3.2498	4.7809	
10	1.3722	1.8125	2.2281	2.7638	3.1693	4.5869	
11	1.3634	1.7959	2.2010	2.7181	3.1058	4.4370	
12	1.3562	1.7823	2.1788	2.6810	3.0545	4.3178	
13	1.3502	1.7709	2.1604	2.6503	3.0123	4.2208	
14	1.3450	1.7613	2.1448	2.6245	2.9768	4.1405	
15	1.3406	1.7531	2.1314	2.6025	2.9467	4.0728	
16	1.3368	1.7459	2.1199	2.5835	2.9208	4.0150	
17	1.3334	1.7396	2.1098	2.5669	2.8982	3.9651	
18	1.3304	1.7341	2.1009	2.5524	2.8784	3.9216	
19	1.3277	1.7291	2.0930	2.5395	2.8609	3.8834	
20	1.3253	1.7247	2.0860	2.5280	2.8453	3.8495	
21	1.3232	1.7207	2.0796	2.5176	2.8314	3.8193	
22	1.3212	1.7171	2.0739	2.5083	2.8188	3.7921	
23	1.3195	1.7139	2.0687	2.4999	2.8073	3.7676	
24	1.3178	1.7109	2.0639	2.4922	2.7969	3.7454	
25	1.3163	1.7081	2.0595	2.4851	2.7874	3.7251	

confidence interval for the mean

- a random variable X in a population is distributed as $N_{\mu,\sigma}(x)$ with μ and σ **unknown**
- a random sample $S := \{x_1, x_2, \dots, x_n\}$ is observed

- **sample mean** $\bar{x} = \frac{1}{n} \sum_{i=1,n} x_i$, **sample standard deviation** $\sigma_S = \sqrt{\frac{\sum_{i=1,n} (x_i - \bar{x})^2}{n-1}}$
UNBIASED
BESSEL'S CORRECTION

it can be shown that $\frac{\bar{x} - \mu}{\frac{\sigma_S}{\sqrt{n}}} \approx t_{n-1}$, i.e. a Student's-t distribution with $\nu = n - 1$ degrees of freedom

$$P\left\{t_{n-1}^{\frac{\alpha}{2}} \leq \frac{\bar{x} - \mu}{\frac{\sigma_S}{\sqrt{n}}} \leq t_{n-1}^{1-\frac{\alpha}{2}}\right\} = 1 - \alpha$$

$$P\left\{\bar{x} - \frac{\sigma_S}{\sqrt{n}} t_{n-1}^{1-\frac{\alpha}{2}} \leq \mu \leq \bar{x} + \frac{\sigma_S}{\sqrt{n}} t_{n-1}^{1-\frac{\alpha}{2}}\right\} = 1 - \alpha$$

if $n > 30$ $P\left\{\bar{x} - \frac{\sigma_S}{\sqrt{n}} Z_{1-\frac{\alpha}{2}} \leq \mu \leq \bar{x} + \frac{\sigma_S}{\sqrt{n}} Z_{1-\frac{\alpha}{2}}\right\} = 1 - \alpha$

example 2.0:

consider the viral load of asymptomatic covid patients in L'Aquila is distributed as a normal random variable with **unknown mean and standard deviation**. The viral load is observed on a sample of 25 asymptomatic patients and the sample mean is 5.23 copies/ml while the sample standard deviation is 0.7. Define an interval of values $[a, b]$ in which the mean of the whole population lies with a 95% probability.

$$\text{DATA: } \begin{cases} \sigma_S = 0.7 \\ \bar{x} = 5.23 \\ n = 25 \\ 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \end{cases}$$

$$P\left(\bar{x} - \frac{\sigma_S}{\sqrt{n}} t_{n-1}^{1-\alpha/2} \leq \mu \leq \bar{x} + \frac{\sigma_S}{\sqrt{n}} t_{n-1}^{1-\alpha/2}\right) = 1 - \alpha$$

$$\text{UNKNOWN: } \begin{cases} a \\ b \end{cases}$$

$$P\left\{5.23 - \frac{0.7}{\sqrt{25}} t_{24}^{0.975} \leq \mu \leq 5.23 + \frac{0.7}{\sqrt{25}} t_{24}^{0.975}\right\} = 0.95$$

16	1.3368	1.7459	2.1199
17	1.3334	1.7396	2.1098
18	1.3304	1.7341	2.1009
19	1.3277	1.7291	2.0930
20	1.3253	1.7247	2.0860
21	1.3232	1.7207	2.0796
22	1.3212	1.7171	2.0739
23	1.3195	1.7139	2.0687
24	1.3178	1.7109	2.0639
25	1.3163	1.7081	2.0595

$$t_{24}^{0.975} = 2.0639, \quad \frac{0.7}{\sqrt{25}} = 0.14$$

$$a = 5.23 - 2.0639 * 0.14 = 4.941$$

$$b = 5.23 + 2.0639 * 0.14 = 5.519$$

$$P\{4.941 \leq \mu \leq 5.519\} = 0.95$$



test of hypothesis

two alternative hypotheses

- H_0 (null hypothesis): a random variable follows a **certain** distribution with **certain** parameters
- H_1 : H_0 is false (different alternatives are possible)
↳ ALTERNATIVE HYPOTHESIS

by observing a sample of the random variable is it possible to say if the random variable follows the distribution of H_0 with a certain confidence level $1 - \alpha$?

- 1 not possible to reject H_0 (different from accepting H_0)
- 2 H_0 is rejected (H_0 is false and H_1 is true)

two settings

- one sample test
- two samples test

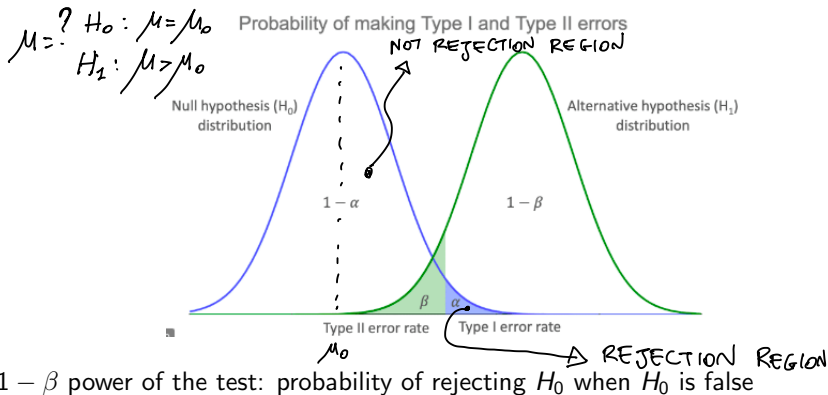
H_1 IS WEAKER THAN H_0
 $\Rightarrow$ WE CAN ACCEPT H_1
WE CANNOT ACCEPT H_0



test of hypothesis

two possible errors

- **type 1** error: rejecting H_0 when H_0 is true (probability α)
- **type 2** error: not rejecting H_0 when H_0 is false (probability β)



one sample hypothesis test

- a random variable X in a population is normally distributed with known standard deviation σ and an hypothesis test is made on the value of its mean μ with confidence level $1 - \alpha$
 - ▶ $H_0: \mu = \mu_0$
 - ▶ $H_1: \mu > \mu_0$ (one sided alternative)
- a random sample $S := \{x_1, x_2, \dots, x_n\}$ is observed with sample mean $\bar{x} = \frac{1}{n} \sum_{i=1, n} x_i$

if H_0 is "true" $\frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \approx N(x)$ then

$$P\left\{\frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} \leq Z_{1-\alpha}\right\} = 1 - \alpha \quad \Rightarrow \quad P\{\bar{x} \leq \mu_0 + \frac{\sigma}{\sqrt{n}} Z_{1-\alpha}\} = 1 - \alpha$$

if $\begin{cases} \bar{x} \leq \mu_0 + \frac{\sigma}{\sqrt{n}} Z_{1-\alpha}, & H_0 \text{ not rejected (error type 2 with probability } \beta) \\ \bar{x} > \mu_0 + \frac{\sigma}{\sqrt{n}} Z_{1-\alpha}, & H_0 \text{ rejected (error type 1 with probability } \alpha) \end{cases}$

example:

a normal random variable has standard deviation $\sigma = 3.6$ with unknown mean. Consider the following hypotheses

- $H_0: \mu = 12$
- $H_1: \mu > 12$

by observing a sample of 16 realizations of the random variable with sample mean 13.7, ~~can we accept~~ H_0 with confidence level 95% ($\alpha = 0.05$)?

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07
+0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790
+0.1	.53983	.54380	.54776	.55172	.55567	.55966	.56360	.56749
+0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642
+0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431
+0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082
+0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566
+0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857
+0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935
+0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785
+0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398
+1	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769
+1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900
+1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796
+1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91308	.91466
+1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922
+1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179
+1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254
+1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164
+1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926
+1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558
+2	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077

→ NOT REJECT

$$12 + \frac{3.6}{\sqrt{16}} Z_{0.95} = 12 + 0.9 * 1.645 =$$

$$= 12 + 1.4805 = 13.4805$$

$$13.7 > 13.4805 \Rightarrow H_0 \text{ rejected}$$

two samples hypothesis test

- two random variables X and Y are normally distributed with the same variance and an hypothesis test is made on their means μ_X, μ_Y with confidence level $1 - \alpha$
 - $H_0: \mu_X = \mu_Y$
 - $H_1: \mu_X \neq \mu_Y$ (two sided alternative)
- random samples $S_X := \{x_1, x_2, \dots, x_{n_X}\}, S_Y := \{y_1, y_2, \dots, y_{n_Y}\}$ are observed with sample means $\bar{x}, \bar{y}$ and sample variances σ_X, σ_Y

it is possible to show that if H_0 is "true" ($\mu_X = \mu_Y$) then

$$\text{TEST STATISTIC } \Psi = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_X^2(n_X-1) + \sigma_Y^2(n_Y-1)}{n_X + n_Y - 2} \left(\frac{1}{n_X} + \frac{1}{n_Y} \right)}} \approx t_{n_X + n_Y - 2}$$

$$P\{t_{n_X + n_Y - 2}^{\frac{\alpha}{2}} \leq \Psi \leq t_{n_X + n_Y - 2}^{1 - \frac{\alpha}{2}}\} = 1 - \alpha$$

$$\text{if } \begin{cases} t_{n_X + n_Y - 2}^{\frac{\alpha}{2}} \leq \Psi \leq t_{n_X + n_Y - 2}^{1 - \frac{\alpha}{2}}, & H_0 \text{ not rejected} \\ \Psi < t_{n_X + n_Y - 2}^{\frac{\alpha}{2}} \text{ or } \Psi > t_{n_X + n_Y - 2}^{1 - \frac{\alpha}{2}}, & H_0 \text{ rejected} \end{cases}$$

example:
SAMPLE RANDOM VARIABLES
two ~~samples~~ X and Y normally distributed with the same variance are observed...

$$\text{DATA: } \begin{cases} \bar{x} = 26.8 \\ \sigma_X = 1.7 \\ n_X = 30 \\ \bar{y} = 27.2 \\ \sigma_Y = 0.8 \\ n_Y = 30 \end{cases}$$

$$\text{HYPOTHESES: } \begin{cases} H_0 : \mu_X = \mu_Y \\ H_1 : \mu_X \neq \mu_Y \\ 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \end{cases}$$

$$\frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_X^2(n_X-1) + \sigma_Y^2(n_Y-1)}{n_X + n_Y - 2} \left(\frac{1}{n_X} + \frac{1}{n_Y} \right)}} = \frac{26.8 - 27.2}{\sqrt{\frac{1.7^2(29) + 0.8^2(29)}{58} \left(\frac{1}{30} + \frac{1}{30} \right)}} = -1.166$$

$$t_{58}^{0.025} \dots$$

$$t_{58}^{0.975} \dots$$



example:

two samples X and Y normally distributed with the same variance are observed...

$$\text{DATA: } \begin{cases} \bar{x} = 26.8 \\ \sigma_X = 1.7 \\ n_X = 30 \\ \bar{y} = 27.2 \\ \sigma_Y = 0.8 \\ n_Y = 30 \end{cases} \quad \text{HYPOTHESES: } \begin{cases} H_0 : \mu_X = \mu_Y \\ H_1 : \mu_X \neq \mu_Y \\ 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05 \end{cases}$$

$$\frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_X^2(n_X-1) + \sigma_Y^2(n_Y-1)}{n_X + n_Y - 2} \left(\frac{1}{n_X} + \frac{1}{n_Y} \right)}} = \frac{26.8 - 27.2}{\sqrt{\frac{1.7^2(29) + 0.8^2(29)}{58} \left(\frac{1}{30} + \frac{1}{30} \right)}} = -1.166$$

$$t_{58}^{0.025} \approx Z_{0.025} = -1.96$$

$$t_{58}^{0.975} \approx Z_{0.975} = 1.96$$

$$-1.96 < -1.166 < 1.96 \Rightarrow H_0 \text{ not rejected}$$

χ^2 test is used for categorical random variables

k STATES IN Ω

DISCRETE

- a random variable E can assume a finite set of k values $E_1, E_2, \dots, E_k$
- according to its probability distribution the expected frequencies of the values are $f_1, f_2, \dots, f_k$ with $f_i \in (0, 1)$

↓

PROBABILITIES

SAMPLE

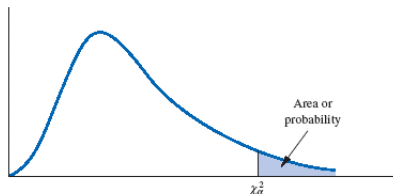
- frequencies $o_1, o_2, \dots, o_k$ are observed on a random sample of a random variable X
- $H_0: X = E$
 $H_1: X \neq E$

it is possible to show that if H_0 is true, then

$$\sum_{i=1, k} \frac{(o_i - f_i)^2}{f_i} \approx \chi_{k-1}^2$$

$\Rightarrow \chi^2$ table can be used to verify H_0

TABLE 11.1 SELECTED VALUES FROM THE CHI-SQUARE DISTRIBUTION TABLE*



Degrees of Freedom	Area in Upper Tail							
	.99	.975	.95	.90	.10	.05	.025	.01
1	.000	.001	.004	.016	2.706	3.841	5.024	6.635
2	.020	.051	.103	.211	4.605	5.991	7.378	9.210
3	.115	.216	.352	.584	6.251	7.815	9.348	11.345
4	.297	.484	.711	1.064	7.779	9.488	11.143	13.277
5	.554	.831	1.145	1.610	9.236	11.070	12.832	15.086
6	.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.647	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.041	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.121	14.611	16.473	34.382	37.652	40.646	44.314

example:

verify if a 6-sided dice is unbiased having observed the outcomes of 600 throws with confidence level 99%.

CONTINGENCY TABLE

	1	2	3	4	5	6
observed	120	87	103	100	104	86
expected	100	100	100	100	100	100

HYPOTHESES:

$$\begin{cases} H_0 : \text{dice is unbiased} \rightarrow f_i = 1/6 \\ H_1 : \text{dice is biased} \\ 1 - \alpha = 0.99 \Rightarrow \alpha = 0.01 \end{cases}$$

$$\begin{aligned} \sum_{i=1,k} \frac{(o_i - f_i)^2}{f_i} &= \frac{(\frac{120-100}{600})^2}{100/600} + \frac{(\frac{87-100}{600})^2}{100/600} + \frac{(\frac{103-100}{600})^2}{100/600} + \frac{(\frac{100-100}{600})^2}{100/600} + \\ &+ \frac{(\frac{104-100}{600})^2}{100/600} + \frac{(\frac{86-100}{600})^2}{100/600} = 0.013167 \end{aligned}$$

$$\chi_5^{20.01} = 15.086 > 0.013167 \Rightarrow H_0 \text{ not rejected}$$